

Furstenberg's Correspondence Principle

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Theorem 0.1 (cf. Kra et al., 2022, Theorem 2.10). *Let Γ be an amenable group, $A \subset \Gamma$, and Φ be a Følner sequence in Γ such that the limit*

$$\delta = \lim_{N \rightarrow \infty} \frac{|A \cap \Phi_N|}{|\Phi_N|}$$

exists.

Then there exists an ergodic system (X, μ, Γ) that is acted on by Γ , a clopen set $E \subset X$, a Følner sequence Ψ in Γ , and a point $a \in X$ that is generic with μ with respect to Ψ such that $\mu(E) \geq \delta$ and

$$A = \{\gamma \in \Gamma : T^\gamma a \in E\}.$$

Theorem 0.2 (cf. Kra et al., 2026, Theorem 2.5). *Let Γ be an amenable group, $A \subset \Gamma$, and Φ be a Følner sequence in Γ such that the limit*

$$\delta = \lim_{N \rightarrow \infty} \frac{|A \cap \Phi_N|}{|\Phi_N|}$$

exists.

Then there exists an ergodic measure-preserving system (X, μ, Γ) that is acted on by Γ , a clopen set $E \subset X$, a Følner sequence Ψ in Γ , and a point $a \in X$ such that:

1. *The point a is generic with μ with respect to Ψ .*
2. *$\mu(E) \geq \delta > 0$.*
3. *$A = \{\gamma \in \Gamma : T^\gamma a \in E\}$.*
4. *Every measurable eigenfunction is equal to μ -almost everywhere to a topological eigenfunction.*

See:

(Kra et al., 2022, Proposition 3.20 and Lemma 5.7):

- Suppose for each $j \in \mathbb{N}$, we have a system (X_j, μ_j, T_j) and suppose there are continuous factor maps $\pi_j : X_j \rightarrow X_{j-1}$ for all $j > 1$. Then there exists

a unique *inverse limit* (X, μ, T) , which is a system admitting continuous factor maps $\psi_k : X \rightarrow X_j$ such that $\pi_j \circ \psi_j = \psi_{j-1}$ for every $j > 1$, and such that

$$\bigcup_{j \in \mathbb{N}} C(X_j) \circ \psi_j$$

is dense in $C(X)$. Additionally, whenever each of the systems (X_j, μ_j, T_j) is ergodic, so is the inverse limit (X, μ, T) .

- According to the original source [Nilpotent Structures in Ergodic Theory], this applies to topological dynamical systems generally.
- Theorem 5.3: Fix an ergodic system (X, μ, T) . For every $k \in \mathbb{N}$, there is a system (Z_k, m_k, T) with all the following properties:
 - The system (Z_k, m_k, T) is an inverse limit of ergodic k -step nilsystems.
 - There is a measurable factor map ρ_k from (X, μ, T) to (Z_k, m_k, T) .
 - For any $f \in L^\infty(X, \mu)$, we have $\|f\|_{k+1} = 0$ if and only if f and $L^\infty(Z_k, m_k) \circ \rho_k$ are orthogonal in $L^2(X, \mu)$.
- Definition 5.4 & 5.5:
 - Whenever (X, μ, T) is ergodic, we call the system (Z_k, m_k, T) satisfying the conclusion of Theorem 5.3 the *k-step pronilfactor* of (X, μ, T) .
 - Given an ergodic system (X, μ, T) and $k \in \mathbb{N}$, let (Z_k, m_k, T) be its k -step pronilfactor. We say that (X, μ, T) *has topological pronilfactors* if, for every $k \in \mathbb{N}$, there is a continuous factor map $\pi_k : X \rightarrow Z_k$.
- Proposition 5.6 is equivalent to Charamaras and Mountakis (2024, Lemma 3.5)
- Lemma 5.7 is a generalisation of Proposition 3.20.
 - Both start with an ergodic system.
 - Instead of $a \in \text{gen}(\mu, \Phi)$, a is a transitive point (admits dense orbit).
 - The systems shown to exist:
 - * Both be ergodic systems with Folner sequences with a generic point and a continuous factor map such that a maps to this generic point.
 - * Have topological pronilfactors in Lemma 5.7, instead of a continuous factor map to its Kronecker factor.
 - Lemma 5.7 also states the original ergodic system and the existing system are measure theoretically isomorphic systems.
 - * Proposition 3.20 states this within its proof
- Proof of Lemma 5.7 vs Proposition 3.20:
 - Measurable Factor Maps: Existence of measurable factor map/s to its pronilfactors or Kronecker Factor.
 - Product/Kronecker system:
 - * Lemma 5.7 creates a product space by taking the countable product over $k \in \mathbb{N}$ of k -step pronilfactors.
 - * Proposition 3.20 focuses solely on the 1-step pronilfactor, i.e., Kronecker factor.
 - * The transformation on these spaces is the product transformation of all k -step pronilfactors and is distal on this product system.
 - * Using a previous proposition, or essentially Charamaras and

Mountakis (2024, Lemma 3.5), to determine the generic point in the product space or Kronecker space.

- New system:
 - * A new system is created by taking the product between the original system and the Product/Kronecker system, defining the generic point as the product of the generic points in each system and the transformation similarly.
 - * The measure is defined as the push forward of the original measure with the measurable factor maps to each coordinate in the new system.
 - * Lemma 5.7 defines the new space to be the orbit closure of the generic point in the constructed product space and defining the projection of the first coordinate onto the original system.
- The continuous map from the new system to the original system is defined as the projection on the first coordinate.
 - * Similarly, the projection on the remaining coordinates are continuous maps from the new system to the original system's pronilfactors.
- Isomorphism: The maps $x \mapsto (x, \rho(x))$ and $(x, z) \mapsto z$ are inverses of each other almost everywhere and so the two spaces are measure theoretically isomorphic.
 - * Thus, the new system is ergodic and the k -step pronilfactors of each system are isomorphic to each other.
- Conclusion: the continuous maps from the new system to the original's k -step pronilfactors and the new system's generic point maps to the generic points in the k -step pronilfactors define continuous factor maps from the new system to its k -step pronilfactors.

(Charamaras and Mountakis, 2024, Lemma 3.5 & Proposition 3.7):

- Lemma 3.5: Covered in Kronecker factor notes and already generalised to amenable groups. It is also a generalisation of Kra et al. (2022, Propositions 3.19 & 5.6).
- Proposition 3.7: Similar generalising of Kra et al. (2022, Proposition 3.20)

(Kra et al., 2025, Definition 4.3 & Subsection 4.4):

- Definition 4.3 (Nilsystems, pronilsystems): Let Γ be an s -step nilpotent Lie group and let $\Gamma \subseteq \Gamma$ be a discrete and cocompact subgroup. The compact manifold $X = \Gamma/\Gamma$ is an s -step nilmanifold.
 - The group Γ acts on X by left translation, and there is a unique Borel probability measure μ on X that is invariant under this action, the *Haar measure* on X . Letting $T : X \rightarrow X$ denote left translation by a fixed element of Γ , the resulting measure preserving system (X, μ, T) is called an s -step nilsystem (as usual endowed with the Borel σ -algebra).
 - An inverse limit of s -step nilsystems is called an s -step pronilsystem.

Notes:

- This result is FCP but skipping the shift space straight to the system that has a continuous factor map to its Kronecker factor.
 - This is where property 4 comes from.

1 Comparison with Functional Analysis Approach

1.1 Defining the Algebra

(Kra and Host, 2007, p. 242) uses a separable subalgebra, $\mathcal{A}$, which is a unitary subalgebra of $\ell^\infty(\mathbb{Z})$ that is invariant under the shift and under complex conjugation, closed in $\ell^\infty(\mathbb{Z})$ and separable for the uniform norm written $\|\cdot\|_\infty$. They have the intent to mostly consider the case of the separable subalgebra $\mathcal{A}(a)$ spanned by a bounded sequence $a = (a_n : n \in \mathbb{Z})$. They also write σ for the shift on $\ell^\infty(\mathbb{Z})$ where, for a sequence $a = (a_n : n \in \mathbb{Z})$, $\sigma a = (a_{n+1} : n \in \mathbb{Z})$ and $\bar{a}$ to denote the conjugate sequence $(\bar{a}_n : n \in \mathbb{Z})$.

This means that:

- We have the constant sequence $1 = (1 : n \in \mathbb{Z}) \in \mathcal{A}$.
- For $f \in \mathcal{A}$, we have $\|f\|_\infty = \sup_{n \in \mathbb{Z}} |f_n|$, $\sigma f \in \mathcal{A}$ and $\bar{f} \in \mathcal{A}$.
- For $f, g \in \mathcal{A}$, we have $f + g \in \mathcal{A}$ and $f \cdot g \in \mathcal{A}$.
- If $f_n \in \mathcal{A}$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} \|f_n - f\|_\infty = 0$, then $f \in \mathcal{A}$.
- (Countable dense subset) There exists a countable set $\{f_n : n \in \mathbb{N}\} \subseteq \mathcal{A}$ such that, for every $f \in \mathcal{A}$ and every $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $\|f_n - f\|_\infty < \varepsilon$.
- $\mathcal{A}(a)$ is generated by applying any combination of the above to a and then taking the closure.

With our approach, the subalgebra is a consequence of our construction. For example, $E \subset \Gamma$ corresponds to a vector where

$$e(\gamma) \begin{cases} 1 & \gamma \in E, \\ 0 & \gamma \notin E. \end{cases}$$

This gives us

$$e = (e(\gamma) : \gamma \in \Gamma) \in \ell^\infty(\Gamma),$$

and $\mathcal{A}(e)$ is the subalgebra we work with, where we define the shift using the group action $T^{\gamma_1} e(\gamma_2) = e(\gamma_1 \cdot \gamma_2)$. Thus, the approach is equivalent so far.

2 Defining the Dynamical System

(Kra and Host, 2007, p. 242) goes on to let X' be the *Gelfand spectrum* of $\mathcal{A}$, stating this means X consists of the set of unitary homomorphisms from $\mathcal{A}$ to the complex numbers.

This means that: - For all $f, g \in \mathcal{A}$, $\lambda \in \mathbb{C}$ and $\varphi \in X'$, we have $\varphi(f + g) = \varphi(f) + \varphi(g)$, $\varphi(f \cdot g) = \varphi(f) \cdot \varphi(g)$, and $\varphi(\lambda f) = \lambda \varphi(f)$. - For all $\varphi \in X$, we have $\varphi(1) = 1$.

They then let $C(X)$ denote the algebra of continuous functions on X and conclude that there exists an isometric isomorphism of algebras $\Phi : C(X) \rightarrow \mathcal{A}$. They also define, for $b \in \mathcal{A}$, the function $\Phi^{-1}(b)$ as *the function associated to b* .

With our approach, we instead use the shift space $X = \{0, 1\}^\Gamma$ such that $e \in X$, which ignores the subalgebra construction, and define a topological dynamical system.

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Kra, B. and Host, B. (2007). *Uniformity seminorms on ℓ^∞ and applications*. Available at: <https://arxiv.org/abs/0711.3637>